



Proposal Presentation

Development and Validation of a Dexterous Hand System

··· Dexhand Designed by The Robot Studio···

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DexHand is an **open-source** dexterous hand for humanoid robotics research, balancing **low cost**, **reproducibility**, and **extensibility** across mechanical design, control, and simulation.

Project Development

Project Origin

Proposed and released as an open-source V1 design by Rob Knight

Hardware Maturity

The IoT Design Shop continuously improved the BOM, assembly wiki, and design iterations, making the robotic hand easier to replicate, wire, and debug.

Control Expansion

In August 2023, BLE firmware and electronics setup documentation were released, supporting the Arduino Nano RP2040 Connect-based control architecture.

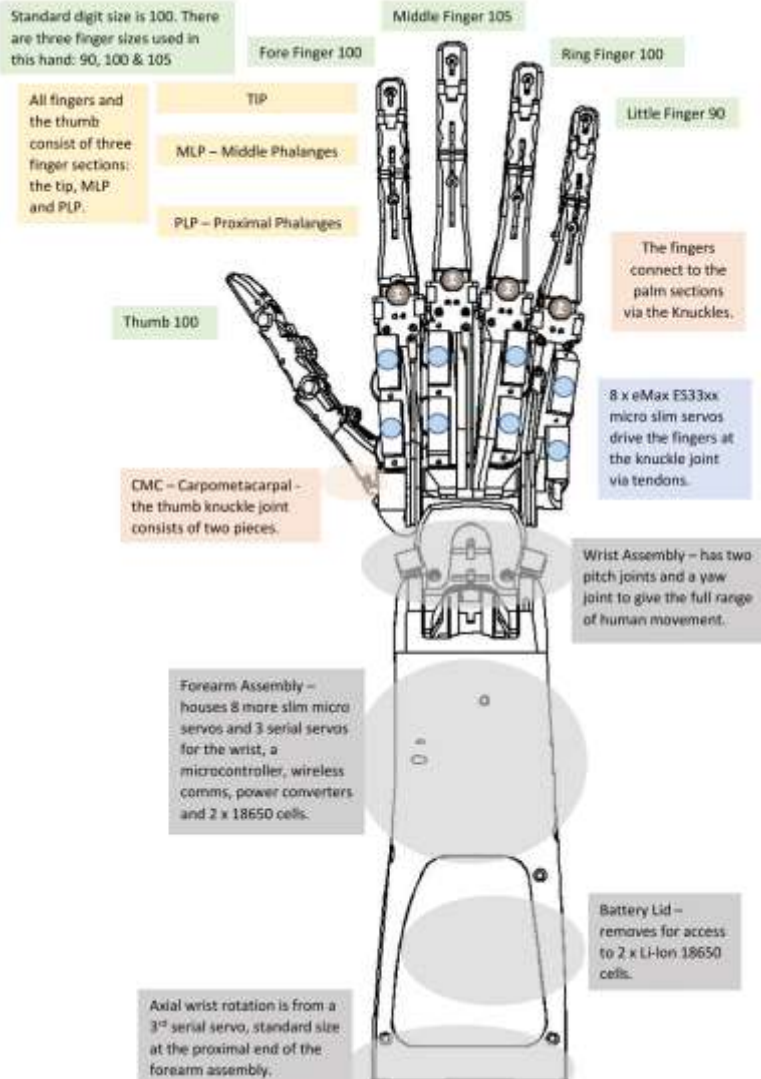
Software Ecosystem

In October 2023, ROS 2 support was introduced, together with URDF and RViz visualization/demo packages, facilitating simulation and secondary development.

Dexhand Whole Body



Gantt Chart (Final)



Finger

Five-finger biomimetic design with three phalanges: TIP, MLP, and PLP.

Wrist

Provides pitch and yaw for more natural motion.

Forearm

Integrates actuators, power supply, and controller.

Servo

Tendon-driven servos enable coordinated multi-joint motion.

Key Structural Features

Structural Parameters

- **Five-finger anthropomorphic layout**
- **3 phalanges per finger**
- **TIP / MLP / PLP joint structure**
- **15 DOF for the fingers**
- **3 DOF for the wrist**
- **Pitch, yaw, and axial wrist rotation**
- **Tendon-driven transmission**
- **Integrated hand–wrist–forearm architecture**



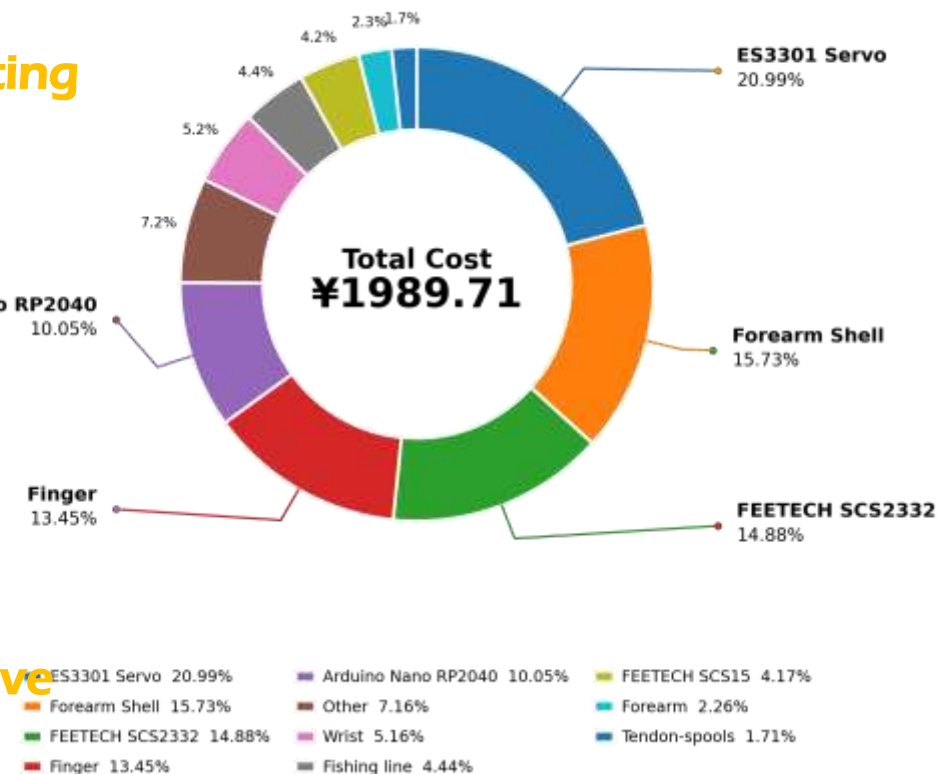
Components		Cost	Ratio	Satus
Finger	Thumb	36.41	1.83%	Preparing
	Index	50.68	2.55%	Preparing
	Middle	81.34	4.09%	Purchased
	Ring	48.41	2.43%	Preparing
	Little	50.77	2.55%	Preparing
Wrist		102.67	5.16%	Preparing
Forearm		45.05	2.26%	Preparing
Forearm Shell		313	15.73%	Preparing
tendon-spools		34	1.71%	Preparing
Arduino Nano RP2040 Connect		1*200=200	10.05%	Purchased
ES3301 Servo		16*26.1=417.6	20.99%	Purchased
FEETECH SCS2332		2*148=296	14.88%	Purchased
FEETECH SCS15		1*83=83	4.17%	Purchased
Fishing line No. 8		1*88.4=88.4	4.44%	Purchased
Other (See the BOM table for details)		142.38	7.16%	Purchased
Total		1989.71		

Other: 4cm*6cm Arduino通用原型板/2.54 排母20pin/2.54 排针20pin/圆柱销套件/M2 螺丝及垫片/螺纹嵌件
 m2*4*3.5/12编100m灰色/6103滚珠轴承/263滚珠轴承/深沟球轴承6702ZZ/693zz 384/m3*60/m3*70

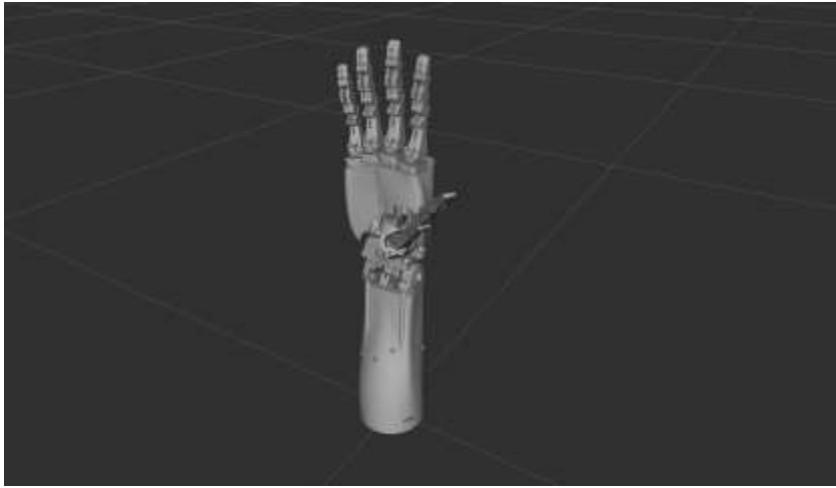
3D Printing

Arduino Nano RP2040

Joint drive



Open-Hand Pose



Closed-Hand Pose



ROS2 Simulation With Rviz2

Completed Work

- Set up the ROS2 simulation environment for DexHand
- Successfully launched the dexterous hand model
- Verified basic joint and finger motion in simulation
- Confirmed the basic visualization and control workflow

Outcome

- Demonstrated that the DexHand model can be launched, visualized, and controlled in ROS2
- Provided a validation basis for later hardware integration



Main Workflow For Simulation

Robot Model

ROS2 Launch and Packages

Visualization and Joint Control

Motion Verification



Hand Model

```
✓ DEXHAND_WS
> build
> install
> log
✓ src
> dexhand_description
> dexhand_gesture_controller
> dexhand_llm_control
> dexhand_usb_serial
```

Workspace Packages



RViz Interface

System Understanding

- Understood the basic ROS2 simulation workflow of DexHand
- Identified the link between robot description, launch files, and motion control

What We Verified

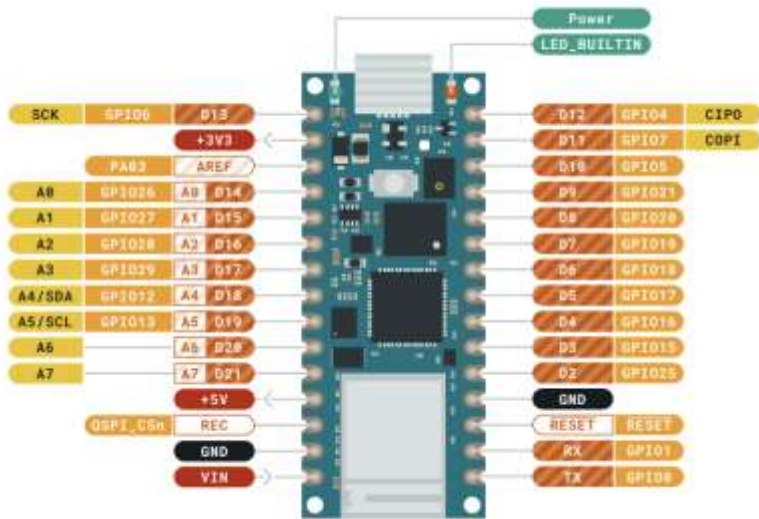
- Correct loading and visualization of the hand model
- Consistency between commanded motion and simulated poses

Why It Matters

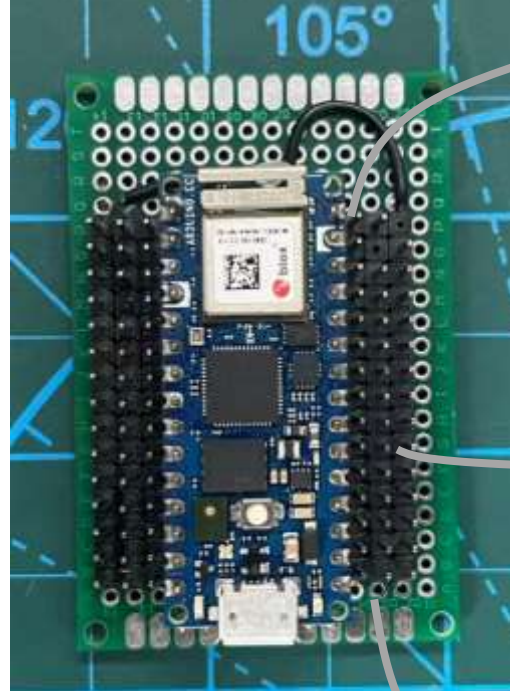
- Reduces uncertainty before physical assembly
- Supports later hardware integration and future AI-based extensions

Main Control

- Based on **Arduino Nano RP2040 Connect**
- Fits inside the forearm
- Controls **all 18 servos**
- Supports **USB serial** and **Bluetooth LE**



- Built from off-the-shelf **Arduino** and protoboard components
- Compact layout designed for forearm integration
- Supports multi-servo wiring and signal distribution



Finger and Thumb Servos × 16



ES3301

Wrist Flexion/Extension × 2



Feetech SCS2332

Axial Wrist Rotation × 1



Feetech SCS15



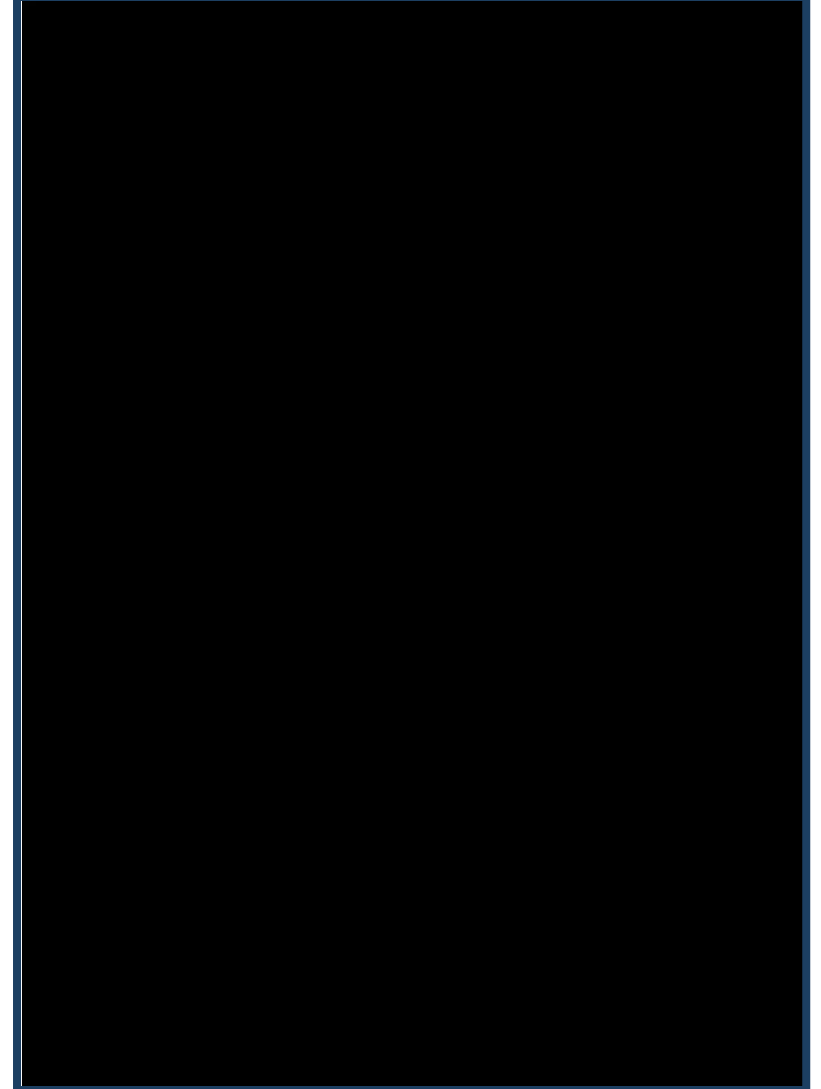
LLM-Enabled Natural Language Control

- **LLM to ROS 2 Bridge:** Acts as a *seamless communication link* between LLM and the DexHand Gesture Controller.
- **Embodied Actuation:** Enables actions where the dexterous hand serves as a physical output medium for AI-generated responses.

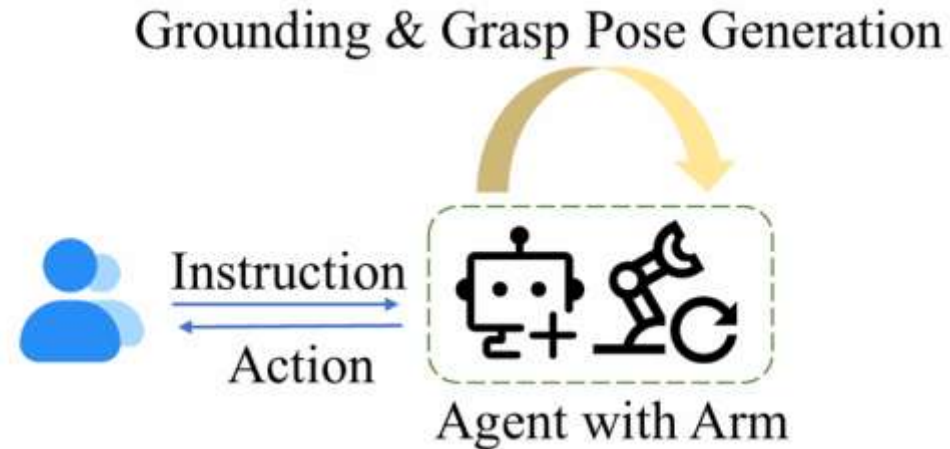
Semantic Task Execution

- **Mathematical Interaction:** LLM can be asked math questions and use hand poses to indicate answers or logic.
- **Gesture Synthesis:** Interprets natural language commands to perform specific gestures, such as a “*Devil Horns*”.

Video Demonstrate



Workflow



How language control achieved

System Architecture and Control Workflow:

- **Speech-to-Text Processing:** SpeechRecognition package and Google Speech Recognition.
- **LLM Intelligence Layer:** Forwards the text to LLM via the API to parse intent and identify specific hand gestures.
- **Message Dispatch:** Sends validated poses or gestures to the DexHand Gesture Controller.

Dynamic Motion Synthesis(optional):

Generalizable Motion Synthesis: Aims to **shift the control logic** away from relying solely on built-in firmware actions and pre-defined poses.



Thank you for listening!

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